

INTRO

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This is a simple applet showcasing the power of the [rumtk](#) framework. The goal of this applet is to provide a simple interface for running different kinds of benchmarks on the framework. I also hope to add an export feature to the applet so the results can be saved to disk for record keeping.

This utility is not specific to the [rumtk](#) toolset. You can configure your own profiles to examine your own programs. This benchmarking tool might not be appropriate for interpreted programs since we look at the binary's call stack. As a result, I suspect that the only useful information you might obtain from benchmarking an interpreted program is statistics about the interpreter itself.

THE BENCHMARK REPORT IS BROKEN DOWN INTO THESE SECTIONS:

- Meta: A small section showing the target architecture and OS of this rumtk-benchmark build.
- File List: list of test files and their sizes in MB used for running test of utility.
- Basic: A simple run of the selected utility using hyperfine. The resulting report shows the mean times with breakdowns by user and kernel spaces.
- Flamegraph: Interactive flamegraph attempting to visualize call graph of the test utility under load. Helpful in identifying function targets of interest for future optimizations. This visualization is generated by performing a quick benchmark with the perf command and then passing the report to [flamegraph-rs](#).
- CPU Summary: Simple summary report of basic CPU statistics obtained with the perf command.
- CPU Cache & Branch Misses: Report of functions showing a % cache and branch miss of > 1%. This helps with identifying if the performance issue is coming from a large number of CPU cache misses. Meaning, if the program is not maximizing the usage of the data already present in the CPU cache lines or a failure of the program to pre-train the CPU algorithm to prefetch necessary data. You do get a technical subreport of the current

WISHLIST:

- Network Summary: Report of the local network information.
- Network Stress Test: Test the interfaces against the message and record information about how fast everything moves through the pipes.

INFORMATION THAT IS NOT COLLECTED:

- 0 Analysis: Currently, we do not gather the information necessary to analyze the general behavior of the program or algorithm. Therefore, this benchmark utility does not provide big O notation information.
- SIMD: Currently, this utility does not inspect and present a disassembly of the test program and in particular the key functions. Thus, no information about proper usage of **SIMD** extensions can be ascertain. I recommend you use Matt Goldbolt's excellent [Compiler Explorer](#) tool or objdump for that.

CONTRIBUTING

If you appreciate the work done in this project, please, visit the following links:

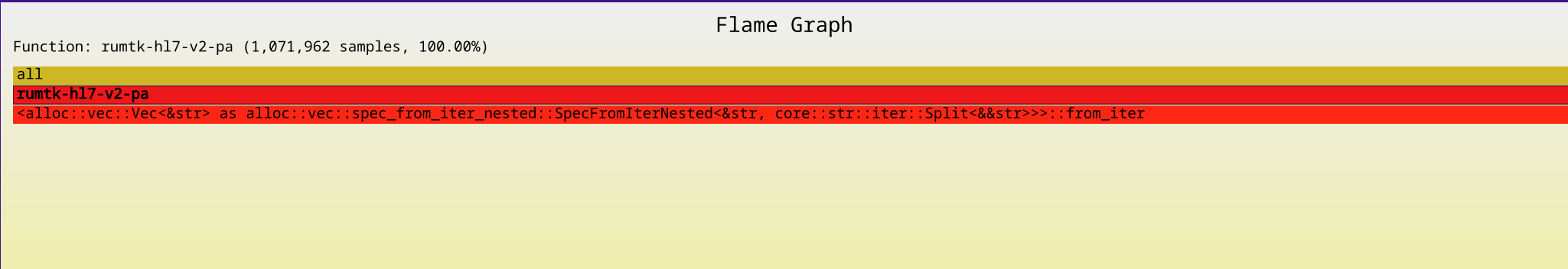
- [OpenCollective](#).
- Navigate to [MedicalMasses L.L.C.'s](#) website.
- Or reach out to [Luis's email](#).

BASIC BENCHMARK

Architecture: x86_64
OS: linux

► Test File Sizes in MB

Command	Mean	Std	Min	Max	Relative	User	Kernel	Runs	Units
`../target/release/rumtk-hl7-v2-parse`	1.1	0.2	0.8	1.8	1	0.3	0.7	2544	[ms]



Performance counter stats for './target/release/rumtk-hl7-v2-parse':

47,167	cache-references:u
13,904	cache-misses:u
1,313,204	cycles:u
1,475,527	instructions:u
300,868	branches:u
122	faults:u
0	migrations:u

0.001956426 seconds time elapsed

0.000000000 seconds user

0.001985000 seconds sys

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captured on : Tue May 5 10:34:04 2026

header version : 1

data offset : 680

data size : 2864

feat offset : 3544

hostname : fedora

os release : 6.19.14-300.fc44.x86_64

perf version : 6.19.14-300.fc44.x86_64

arch : x86_64

nrcpus online : 16

nrcpus avail : 16

cpudesc : AMD Ryzen 7 7730U with Radeon Graphics

cpuid : AuthenticAMD,25,80,0

total memory : 39942832 kB

cmdline : /usr/bin/perf

event : name = cache-misses:u, , id = { 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224 }, type = 4 (cpu), size = 144, config = 0x964 (cache-misses), { sample_period, sample_freq } = 4000, sample_type = IP|TID|TIME|ID|PERIOD, read_format = TOTAL_TIME_ENABLED|TOTAL_TIME_RUNNING|ID|LOST, disabled = 1, inherit = 1, exclude_kernel = 1, exclude_hv = 1, mmap = 1, comm = 1, freq = 1, inherit_stat = 1, enable_on_exec = 1, task = 1, sample_id_all = 1, mmap2 = 1, comm_exec = 1, ksymbol = 1, bpf_event = 1, build_id = 1

event : name = branch-misses:u, , id = { 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240 }, type = 4 (cpu), size = 144, config = 0xc3 (ex_ret_brn_misp), { sample_period, sample_freq } = 4000, sample_type = IP|TID|TIME|ID|PERIOD, read_format = TOTAL_TIME_ENABLED|TOTAL_TIME_RUNNING|ID|LOST, disabled = 1, inherit = 1, exclude_kernel = 1, exclude_hv = 1, freq = 1, inherit_stat = 1, enable_on_exec = 1, sample_id_all = 1

sibling sockets : 0-15

sibling dies : 0-15

sibling threads : 0-1

sibling threads : 2-3

sibling threads : 4-5

sibling threads : 6-7

sibling threads : 8-9

sibling threads : 10-11

sibling threads : 12-13

sibling threads : 14-15

CPU 0: Core ID 0, Die ID 0, Socket ID 0

CPU 1: Core ID 0, Die ID 0, Socket ID 0

CPU 2: Core ID 1, Die ID 0, Socket ID 0

CPU 3: Core ID 1, Die ID 0, Socket ID 0

CPU 4: Core ID 2, Die ID 0, Socket ID 0

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# CPU 5: Core ID 2, Die ID 0, Socket ID 0
# CPU 6: Core ID 3, Die ID 0, Socket ID 0
# CPU 7: Core ID 3, Die ID 0, Socket ID 0
# CPU 8: Core ID 4, Die ID 0, Socket ID 0
# CPU 9: Core ID 4, Die ID 0, Socket ID 0
# CPU 10: Core ID 5, Die ID 0, Socket ID 0
# CPU 11: Core ID 5, Die ID 0, Socket ID 0
# CPU 12: Core ID 6, Die ID 0, Socket ID 0
# CPU 13: Core ID 6, Die ID 0, Socket ID 0
# CPU 14: Core ID 7, Die ID 0, Socket ID 0
# CPU 15: Core ID 7, Die ID 0, Socket ID 0
# node0 meminfo : total = 15444316 kB, free = 427728 kB
# node0 cpu list : 0-15
# node1 meminfo : total = 24498516 kB, free = 14303460 kB
# node1 cpu list : -1
# pmu mappings: cpu = 4, amd_df = 12, amd_iommu_0 = 14, amd_l3 = 13, breakpoint = 5, drm_amdgpu = 4294836224, hwmon_ac = 4294901760, hwmon_acpitz = 4294901761, hwmon_amdgpu =
4294901764, hwmon_bat0 = 4294901762, hwmon_k10temp = 4294901765, hwmon_nvme = 4294901763, hwmon_thinkpad = 4294901766, ibs_fetch = 10, ibs_op = 11, kprobe = 8, msr = 15, power =
16, power_core = 17, software = 1, tool = 4294967294, tracepoint = 2, uprobe = 9
# CPU cache info:
# L1 Data 32K [0-1]
# L1 Instruction 32K [0-1]
# L1 Data 32K [2-3]
# L1 Instruction 32K [2-3]
# L1 Data 32K [4-5]
# L1 Instruction 32K [4-5]
# L1 Data 32K [6-7]
# L1 Instruction 32K [6-7]
# L1 Data 32K [8-9]
# L1 Instruction 32K [8-9]
# L1 Data 32K [10-11]
# L1 Instruction 32K [10-11]
# L1 Data 32K [12-13]
# L1 Instruction 32K [12-13]
# L1 Data 32K [14-15]
# L1 Instruction 32K [14-15]
# L2 Unified 512K [0-1]
# L2 Unified 512K [2-3]
# L2 Unified 512K [4-5]
# L2 Unified 512K [6-7]
# L2 Unified 512K [8-9]
# L2 Unified 512K [10-11]
# L2 Unified 512K [12-13]
# L2 Unified 512K [14-15]
# L3 Unified 16384K [0-15]
# time of first sample : 68071.770606
# time of last sample : 68071.771564
# sample duration : 0.958 ms
# memory nodes (nr 2, block size 0x8000000):
# 0 [15G]: 198-317
# 1 [24G]: 0-25,32-197
# bpf_prog_info empty
# btf info empty
# cpu pmu capabilities: max_precise=0
# AMD systems uses ibs_op// PMU for some precise events, e.g.: cycles:p, see the 'perf list' man page for further details.
# missing features: TRACING_DATA BUILD_ID BRANCH_STACK GROUP_DESC AUXTRACE STAT CLOCKID DIR_FORMAT COMPRESSED CLOCK_DATA HYBRID_TOPOLOGY
# =====
#
#
# Total Lost Samples: 0
#
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# Samples: 8 of event 'cache-misses:u'
# Event count (approx.): 7947
#
# Overhead Command Shared Object Symbol
# .....
#
69.98% rumtk-hl7-v2-pa /home/kiseitai2/RustroverProjects/rumtk/target/release/rumtk-hl7-v2-parse 0x11a3c4 B [.] std::env::args_os
25.47% rumtk-hl7-v2-pa /usr/lib64/ld-linux-x86-64.so.2 0x143a3 B [.] _dl_check_map_versions
4.11% rumtk-hl7-v2-pa /usr/lib64/ld-linux-x86-64.so.2 0x17231 B [.] update_active.constprop.0

# Samples: 10 of event 'branch-misses:u'
# Event count (approx.): 9237
#
# Overhead Command Shared Object Symbol
# .....
#
28.25% rumtk-hl7-v2-pa /usr/lib64/libc.so.6 0x83ee3 B [.] malloc
25.82% rumtk-hl7-v2-pa /usr/lib64/libc.so.6 0x14b060 B [.] __memcmp_avx2_movbe
24.74% rumtk-hl7-v2-pa /home/kiseitai2/RustroverProjects/rumtk/target/release/rumtk-hl7-v2-parse 0x10c39e B [.] <std::sys::sync::once::futex::Once>::call
15.83% rumtk-hl7-v2-pa [unknown] 0xffffffff86a01cf0 ! [k] 0xffffffff86a01cf0
4.89% rumtk-hl7-v2-pa /usr/lib64/ld-linux-x86-64.so.2 0x26e04 B [.] strcmp

#
# (Tip: When collecting LBR backtraces use --stitch-lbr to handle more than 32 deep entries: perf record --call-graph lbr ; perf report --stitch-lbr)
#

```